Electronics Engineer Build Guide

Smart Medication Dispenser

SMD-100 Home Device • SMD-200 Travel Device

VERSION	2.0
DATE	February 2026
CLASSIFICATION	Confidential — Engineering Team
SOURCE	Technical Documentation v4.0
GPIO AUTHORITY	Firmware `pins.h`
COMPLIANCE	CE Class IIa, Swissmedic
LOCATION	Lausanne, Switzerland
TOTAL BUILD TIME	~8 hours (SMD-100)

STEP-BY-STEP ASSEMBLY • TESTING • CALIBRATION

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PHASE 0

Before You Start — Prototyping Roadmap

Planning Phase

Do NOT jump straight to a custom PCB. Follow this phased approach to save time and money.

PHASE	DURATION	GOAL	DELIVERABLE
1. Breadboard	2 weeks	Prove concept	Working motor + sensor on breadboard
2. Dev Boards	4 weeks	Firmware dev	Complete software on dev kits
3. Custom PCB v1	4 weeks	Integration test	First PCB prototype

PHASE	DURATION	GOAL	DELIVERABLE
4. Custom PCB v2	2 weeks	Fix issues	Production-ready PCB
5. Pilot Build	4 weeks	DFM validation	50 units
6. Production	Ongoing	Scale	Mass production

Recommended Dev Boards (Phase 1–2)

BOARD	MCU	PRICE	USE FOR
ESP32-S3-DevKitC-1	ESP32-S3	$10	Firmware, sensors, motor testing
ESP32-S3-LCD-EV-Board	ESP32-S3 + 4.3" LCD	$60	Display/UI with LVGL
WT32-SC01 Plus	ESP32-S3 + 3.5" LCD	$35	Compact UI testing

PHASE 1

Workspace Setup & Equipment

1 day

1.1 — ESD-Safe Workstation

- [ ] Install ESD mat on bench
- [ ] ESD wrist strap connected to ground — wear at ALL times
- [ ] Ground all equipment
- [ ] Maintain 40–60% RH humidity

1.2 — Essential Equipment

#	TOOL	MODEL	~PRICE	PURPOSE
1	Soldering station	Hakko FX-888D	$120	All soldering
2	Hot air rework	Quick 861DW	$150	QFN packages
3	Multimeter	Fluke 117	$200	Voltage/continuity
4	Oscilloscope (4ch, 50MHz+)	Rigol DS1054Z	$350	Signal debug
5	Bench power supply	Korad KA3005D	$80	Controlled power (0-30V)
6	Logic analyzer	Saleae Logic 8	$500	I2C/SPI/UART debug
7	Stereo microscope	AmScope SM-4T	$250	QFN/FFC inspection

1.3 — Consumables

- [ ] Fine ESD tweezers
- [ ] Flush cutters
- [ ] Wire strippers (22-30 AWG)
- [ ] Solder wick
- [ ] Solder paste (SAC305)
- [ ] Solder wire 0.5mm
- [ ] Flux pen (no-clean)
- [ ] IPA 99% + lint-free wipes
- [ ] Kapton tape
- [ ] Heat shrink tubing (2-10mm)
- [ ] Calibration weights: 10g, 50g, 100g
- [ ] Copper tape (EMI shielding)

1.4 — Software Environment

- [ ] Install **ESP-IDF v5.1+** (official Espressif SDK)
- [ ] Install **VS Code** + ESP-IDF extension
- [ ] `esptool.py` (comes with ESP-IDF)
- [ ] Serial terminal (PuTTY / minicom / ESP-IDF monitor)
- [ ] Obtain firmware binary from firmware team
- [ ] USB-to-UART adapter (CP2102 or CH340)

PHASE 2

Component Procurement & Inspection

1–3 weeks

2.1 — ICs & Modules ORDER FIRST

REF	COMPONENT	PART NUMBER	QTY	COST	SUPPLIER
U1	MCU Module	ESP32-S3-WROOM-1-N16R8	1	$5.50	DigiKey
U2	Power Path IC	BQ24195RGER (TI)	1	$3.50	Mouser
U3	5V Buck Regulator	TPS62150RGTR (TI)	1	$1.80	DigiKey
U4	3.3V LDO	AP2112K-3.3TRG1	2	$0.50	LCSC
U5-11	Motor Drivers	ULN2003AN	7	$2.10	LCSC
U7	Audio Amp	MAX98357AETE+	1	$2.50	DigiKey
U8	Load Cell ADC	HX711	1	$0.80	LCSC
U9	Temp/Humidity	SHT40-AD1B	1	$2.50	DigiKey
U10	Light Sensor	BH1750FVI	1	$1.00	LCSC
U11	I/O Expander	MCP23017-E/SO	1	$1.50	DigiKey

2.2 — Display, Motors, Sensors

REF	COMPONENT	SPECIFICATION	QTY	COST	SUPPLIER
LCD1	4.3" TFT Display	Waveshare RGB, 800×480, GT911 touch	1	$28.00	Waveshare
M1-10	Stepper Motors	28BYJ-48 5V	10	$15.00	AliExpress
M11	Gate Servo	SG90 (TowerPro)	1	$2.00	Amazon
LS1	Speaker	40mm, 8Ω, 2W	1	$1.50	DigiKey
Q1-10	Optical Sensors	TCPT1300X01 (Vishay)	10	$8.00	DigiKey
Q11-20	Hall Sensors	A3144	10	$3.00	LCSC
PIR1	Motion Sensor	AM312 mini PIR	1	$1.50	AliExpress
SW1	Reed Switch	NO + neodymium magnet	1	$0.30	LCSC
LC1	Load Cell	1kg TAL220	1	$3.00	Amazon

2.3 — Power & Battery

COMPONENT	PART	QTY	COST	NOTES
Battery Cells	Samsung INR18650-25R	2	$8.00	2500mAh, 20A — buy genuine only!
Battery Holder	Keystone 1042 (2S)	1	$2.00	Series = 7.4V
BMS Board	2S 7.4V 5A (DW01A)	1	$1.50	Over/under voltage + short protection
AC Adapter	Mean Well GST25E12-P1J	1	$12.00	12V/2A — **Swiss Type J plug** + EU adapter

Total BOM Cost: ~$159 per unit (prototype)

2.4 — Incoming Inspection

- [ ] Verify ALL part numbers match order
- [ ] Inspect IC packages for damage (bent pins, cracks)
- [ ] ESP32-S3 module marking must read **"N16R8"** (16MB flash + 8MB PSRAM)
- [ ] Battery cells must be **genuine Samsung** (holographic label, ~45g each)
- [ ] AC adapter: measure 12V ±5% with multimeter
- [ ] Store moisture-sensitive ICs in dry cabinet or sealed bag with desiccant

PHASE 5

Power Section — Build & Verify

1 hour

Build and test power FIRST — before installing the ESP32. If there's a voltage issue, you will destroy the $5.50 MCU module.

5.1 — BQ24195 Power Path Controller QFN-24, 4×4MM

1. Apply solder paste to all 24 QFN pads **+ center thermal pad**
2. Thermal pad is the heat sink — must have paste and thermal vias beneath
3. Place BQ24195RGER with tweezers, align pin 1 marking
4. Verify alignment under microscope
5. Hot air reflow: **260°C peak, 30 seconds**
6. IC "settles" when solder melts — watch edges
7. Inspect ALL pins under microscope — no bridges, no opens
8. Install passives: 10µF (VIN), 22µF (SYS, BAT), NTC resistor

5.2 — TPS62150 Buck Regulator QFN-16, 3×3MM → 5V Rail

1. Apply solder paste to QFN-16 pads + ground pad
2. Place TPS62150RGTR, align pin 1
3. Hot air reflow at 260°C, 20–30 sec
4. Install **2.2µH shielded inductor** close by
5. Install input cap (10µF) and output cap (22µF)

5.3 — AP2112K LDO (SOT-25) → 3.3V Rail

1. Hand solder 5-pin SOT-25 package
2. Install input cap (1µF) + output cap (10µF)
3. Install ferrite bead (600Ω @ 100MHz) on input

5.4 — Power Connectors

1. Solder 12V barrel jack (5.5×2.1mm, center positive)
2. Solder battery connector (JST-XH 2-pin)
3. Install TVS diode for overvoltage protection

5.5 — POWER TEST — Critical Gate

DO NOT proceed until ALL tests below pass.

#	TEST	EXPECTED	MEASURED	PASS?
1	Apply 12V, check current	<10mA quiescent		☐
2	SYS output (BQ24195)	~12V		☐
3	**5V rail** (TPS62150)	**5.0V ±2%**		☐
4	**3.3V rail** (AP2112K)	**3.3V ±2%**		☐
5	No excessive heat	ICs cool after 30s	—	☐

PHASE 6

ESP32-S3 MCU Assembly & First Boot

30 minutes

6.1 — Solder ESP32-S3-WROOM-1-N16R8

1. Apply flux generously to all module pads
2. Apply thin solder paste layer (stencil method preferred)
3. Place module carefully with tweezers
4. Check alignment under microscope — pin 1 marker must match
5. **Preheat board** to 150°C on hot plate
6. Hot air reflow: **260°C for 20–30 seconds**
7. Let cool naturally — do NOT blow cold air
8. Inspect ALL pins under microscope

6.2 — Critical Support Components

- [ ] **EN pin:** 10K pull-up to 3.3V + 0.1µF cap to GND
- [ ] **GPIO0 (BOOT):** 10K pull-up + button to GND
- [ ] **Reset button:** Tactile button on EN → GND
- [ ] **Decoupling:** 10µF + 0.1µF within 3mm of each VCC pin
- [ ] **USB:** GPIO20 (D-) + GPIO21 (D+) to USB connector

6.3 — Antenna Rules

15mm clearance around antenna area. No ground plane underneath. No traces crossing antenna feed. No metal enclosure parts within 10mm.

6.4 — First Boot Verification

1. Connect USB, open serial terminal at **115200 baud**
2. Apply power → should see ESP32 boot log

3. Run `esptool.py flash_id` → confirms **16MB flash**
4. Boot log should show **"8MB PSRAM initialized"**

PHASE 7

Display Connection & Test

30 minutes

Two separate I2C buses:
- **Bus 0 (Touch):** GPIO38 (SDA) + GPIO39 (SCL) — GT911 at 0x5D
- **Bus 1 (Sensors):** GPIO44 (SDA) + GPIO46 (SCL) — SHT40, BH1750, MCP23017

7.1 — Display Wiring (from firmware `pins.h`)

GPIO	SIGNAL	GPIO	SIGNAL	GPIO	SIGNAL
1–5	LCD_R0–R4	6–11	LCD_G0–G5	12–16	LCD_B0–B4
45	**LCD_CLK**	**48**	**LCD_DE**	**47**	**LCD_VSYNC**
21	**LCD_HSYNC**	42	Backlight PWM	—	—
38	TOUCH_SDA	39	TOUCH_SCL	40	TOUCH_INT

PHASE 8

Audio Section

20 minutes

8.1 — I2S Audio Wiring

ESP32 GPIO	MAX98357A	FUNCTION
GPIO17	BCLK	I2S bit clock
GPIO18	LRCLK	I2S L/R clock
GPIO15	DIN	I2S audio data
—	GAIN → GND	15dB gain
—	SD → 3.3V	Enable (always on)
—	VDD → 5V	Power

PHASE 9

Motor Driver Section

30 minutes

9.1 — MCP23017 I/O Expander (address 0x20)

1. Solder SOIC-28 package
2. A0=A1=A2=GND → address **0x20**
3. 100nF decoupling on VDD
4. Connect to I2C Bus 1: SDA → GPIO44, SCL → GPIO46
5. I2C scan should find device at 0x20

9.2 — ULN2003A Motor Drivers (×7)

- [ ] Solder 7× ULN2003AN (SOIC-16 or DIP-16)
- [ ] **COM pin** → **5V** (flyback protection — critical!)
- [ ] Inputs from MCP23017, outputs to motor connectors

- [ ] Motor wire order: Blue-Pink-Yellow-Orange (Red = 5V common)
- [ ] Gate servo on **GPIO43**

> **Motor EMI mitigation:** Add 100nF ceramic cap across each motor coil. Use ferrite beads on motor power lines. Keep motor ground return separate from digital signals.

PHASE 10

Sensor Section

45 minutes

10.1 — I2C Sensors (Bus 1: GPIO44/46)

SENSOR	ADDRESS	PACKAGE	SOLDERING	DECOUPLING
SHT40 (Temp/Humidity)	0x44	DFN 1.5×1.5mm	Hot air	100nF on VDD
BH1750 (Light)	0x23	WSOF	Hot air	100nF on VDD

10.2 — Load Cell + HX711

CONNECTION	GPIO / PIN
HX711 DOUT	**GPIO32**
HX711 SCK	**GPIO33**
Load cell Red (E+)	HX711 E+
Load cell Black (E-)	HX711 E-
Load cell White (A-)	HX711 A-
Load cell Green (A+)	HX711 A+

10.3 — Digital Sensors

SENSOR	GPIO	LOGIC
PIR Motion (AM312)	**GPIO26**	HIGH = motion detected
Reed Switch (Door)	**GPIO27** + 10K pull-up	LOW = closed, HIGH = open
Optical Sensors ×8	MCP23017 GPB0-7	LOW = pill passing
Optical Sensors ×2	GPIO28, GPIO29	LOW = pill passing
Hall Sensors ×10	MCP23017 GPIOs	Triggers at home position

10.4 — SD Card (Phase 11)

GPIO	SD SIGNAL
GPIO36	SD_CLK
GPIO35	SD_CMD (MOSI)
GPIO37	SD_DATA0 (MISO)

PHASES 12–14

Mechanical Assembly, Battery & Enclosure

~2 hours

12 — Mechanical Sub-Assembly

- [ ] Mount 10× stepper motors to bracket (M3 screws, 2 per motor)
- [ ] Press-fit gears onto D-shafts, assemble carousel (10 slots)
- [ ] Mount SG90 gate servo — test full open/close
- [ ] Mount 10× TCPT1300 optical sensors in pill chutes (3mm gap)
- [ ] Mount 10× A3144 Hall sensors near gear teeth + magnets at home positions
- [ ] Mount load cell (TAL220): one end fixed to frame, tray on floating end

13 — Battery & BMS

BMS PROTECTION	THRESHOLD
Overvoltage cutoff	8.4V (4.2V/cell)
Undervoltage cutoff	6.0V (3.0V/cell)
Overcurrent cutoff	5A
Short circuit	Instant cutoff

- [ ] Insert 2× Samsung 18650 cells in holder (series = 7.4V)
- [ ] **Verify polarity before connecting!**
- [ ] Connect BMS board between cells and output
- [ ] Secure battery with foam tape — **NEVER screws near battery**

14 — Final Assembly

1. Install display with foam gasket — connect FFC cable (check orientation!)
2. Mount main PCB on M3 standoffs (10mm)
3. Mount motor/carousel assembly, connect all cables
4. Mount speaker (hot glue/foam), PIR sensor, reed switch
5. Connect battery pack, barrel jack, SD card

6. Pre-close inspection: no pinched wires, antenna clear, all connectors seated
7. Close enclosure (don't overtighten — ABS cracks)

PHASE 15

Firmware Flashing

30 minutes

```
# ESP-IDF
cd %USERPROFILE%\esp\smart_dispenser
idf.py set-target esp32s3
idf.py build
idf.py -p COM3 flash monitor

# Direct flash
esptool.py --chip esp32s3 --port COM3 --baud 921600 write_flash \
  0x0 bootloader.bin  0x10000 firmware.bin  0x8000 partition-table.bin
```

Boot Verification Checklist

- [ ] ESP32-S3 identified in boot log
- [ ] 16MB Flash detected
- [ ] 8MB PSRAM initialized
- [ ] FreeRTOS tasks started
- [ ] LVGL display initialized
- [ ] I2C devices found (0x20, 0x23, 0x44, 0x5D)
- [ ] WiFi connects in <30 seconds
- [ ] API heartbeat sent to server
- [ ] Device serial number programmed (SMD-XXXXXXXX)

PHASE 16

Full Testing & Calibration

1–2 hours

16.1 — Power Tests

TEST	EXPECTED	MEASURED	✓
5V rail under load	5.0V ±2%		☐

TEST	EXPECTED	MEASURED	✓
3.3V rail under load	3.3V ±2%		☐
AC → Battery switchover	<100ms		☐
Battery backup (idle)	>48 hours		☐

16.2 — Component Tests

SYSTEM	TEST	EXPECTED	✓
Display	White/black/color screens	All pixels OK	☐
Touch	4 corners + center	<5mm accuracy	☐
Motors 1–10	Full rotation	Smooth, no stall	☐
Gate servo	Open (90°) / Close (0°)	Full seal	☐
SHT40	Temperature reading	±2°C	☐
BH1750	Light reading	Changes with light	☐
PIR (AM312)	Motion at 1m	Triggers	☐
Reed switch	Magnet near/far	LOW/HIGH correct	☐
Optical ×10	Pass pills through	All count correctly	☐
Hall ×10	Motor homing	All find home	☐
Audio	1kHz tone	Clear, no distortion	☐
SD card	Read/write file	Success	☐
WiFi	Connect 2.4GHz	<30 seconds	☐
API	Heartbeat	Received by server	☐

16.3 — Calibration Procedures

Load Cell Calibration

1. Empty tray → press "Zero" → records offset
2. Place **50g weight** → press "Calibrate" → records scale factor
3. Verify: shows **50.0g ±1g**
4. 10g weight → should read **10.0g ±0.5g**
5. Save calibration to NVS

Motor Homing Calibration

1. For each motor (1–10): run "Find Home" → rotates until Hall sensor triggers
2. Position saved as home (slot 1)
3. Verify: command to any slot → stops at correct position

16.4 — Full Dispense Integration Test

Run `dispense 1 2` (slot 1, 2 pills) and verify the full sequence: carousel rotates → Hall confirms position → gate opens → optical counts 2 pills → gate closes → load cell verifies weight → DOSE_DISPENSED sent to API. **Repeat for all 10 slots.**

PHASE 17

SMD-200 Travel Device Build

2–3 hours

17.1 — Travel Device GPIO Map

GPIO	FUNCTION	DIR	NOTES
0	BOOT button	In	Pull-up
1	I2C SDA	Bidir	OLED + sensors
2	I2C SCL	Out	Shared bus
3	**Cellular TXD**	Out	→ SIM7080G RXD
4	**Cellular RXD**	In	← SIM7080G TXD
5	**Cellular PWRKEY**	Out	Module power control
6	**Cellular STATUS**	In	Module status
7	Buzzer	Out	PWM
8	Vibration motor	Out	On/Off
9–12	Servo PWM 1–4	Out	Gate control
13–16	Optical sensors 1–4	In	Pill counting
17, 18	Buttons (Confirm/Cancel)	In	Pull-up
19	Battery ADC	ADC	Fuel gauge backup
20–21	USB D-/D+	Bidir	Programming

17.2 — SIM7080G Installation

Most challenging component. Large LGA package (17.6×15.7mm). Hot air at 270°C. Inspect carefully for bridges. Attach u.FL antenna cables LAST.

17.3 — Travel Device Test

TEST	EXPECTED	✓
USB-C charging	LED lights, charges	☐
OLED display	128×64 UI shown	☐
LTE Cat-M1 connection	Registers <30s	☐

TEST	EXPECTED	✓
GPS fix	Coordinates received	☐
Servos 1–4	All open/close	☐
Optical sensors 1–4	All detect pills	☐
Buzzer + vibration	Both working	☐
API heartbeat via LTE	Received by server	☐

PHASE 18

Production Testing & Quality Sign-Off

18.1 — In-Circuit Test (ICT)

TEST POINT	EXPECTED	MIN	MAX	MEASURED	✓
VIN (12V)	12.0V	11.4V	12.6V		☐
5V rail	5.0V	4.75V	5.25V		☐
3.3V rail	3.3V	3.2V	3.4V		☐
Battery	7.4V	6.0V	8.4V		☐
ESP32 VCC	3.3V	3.2V	3.4V		☐

18.2 — 24-Hour Burn-In

PHASE	DURATION	CONDITIONS	MONITORING
Normal	8 hours	25°C room temp	IC temps, power draw
High temp	8 hours	40°C chamber	IC temps <60°C
Cycling	8 hours	Dispense every 10 min	100% accuracy

18.3 — Quality Sign-Off

ROLE	NAME	SIGNATURE	DATE
Electronics Engineer			
Firmware Engineer			
QA Inspector			

APPENDIX A

Master GPIO Reference — SMD-100

GPIO	FUNCTION	DIR	BUS	NOTES
0	BOOT / Confirm btn	In	—	10K pull-up
1–5	LCD R0–R4	Out	RGB	Red channel
6–11	LCD G0–G5	Out	RGB	Green channel
12–16	LCD B0–B4	Out	RGB	Blue channel
15	I2S DOUT *(shared B3)*	Out	I2S	Audio data
17	I2S BCLK	Out	I2S	Audio bit clock
18	I2S LRCLK	Out	I2S	Audio L/R clock
20–21	USB D-/D+	Bidir	USB	OTG
26	PIR motion	In	Digital	AM312
27	Reed switch	In	Digital	10K pull-up
32	HX711 DOUT	In	Serial	Load cell data
33	HX711 SCK	Out	Serial	Load cell clock
35	SD CMD (MOSI)	Bidir	SPI	SD card
36	SD CLK	Out	SPI	SD card
37	SD DATA0 (MISO)	Bidir	SPI	SD card
38	Touch SDA	Bidir	**I2C Bus 0**	GT911 @ 0x5D
39	Touch SCL	Out	**I2C Bus 0**	GT911
40	Touch INT	In	Digital	Interrupt
41	Touch RST	Out	Digital	Reset
42	Backlight PWM	Out	LEDC	Via MOSFET
43	Gate servo	Out	LEDC	SG90 PWM
44	Sensor I2C SDA	Bidir	**I2C Bus 1**	SHT40, BH1750, MCP23017
45	LCD CLK	Out	RGB	Pixel clock
46	Sensor I2C SCL	Out	**I2C Bus 1**	Sensor bus
47	LCD VSYNC	Out	RGB	Vertical sync
48	LCD DE	Out	RGB	Data enable

APPENDIX B

Power Consumption Budget

COMPONENT	ACTIVE	SLEEP	NOTES
ESP32-S3	240mA	10µA	WiFi TX peak
Display (4.3" TFT)	150mA	0mA	Backlight 50%
Motors (1 active)	200mA	0mA	During dispense
Sensors	20mA	5mA	I2C polling
Audio (MAX98357A)	300mA	0mA	Peak alert
Other (LEDs, misc)	30mA	5mA	—
TOTAL	**940mA**	**20mA**	**—**

Battery life: Standby: 5000mAh ÷ 20mA = 250 hrs ≈ 10 days. With 4 dispenses/day: ~48 hours.

APPENDIX C

Common Pitfalls & Solutions

PITFALL	PROBLEM	SOLUTION
Motor EMI	Stepper causes WiFi interference	100nF across each coil + ferrite beads
Power brown-out	ESP32 resets on motor start	Inrush current limiting, adequate supply
Touch erratic	Random touch events	Ground display bezel, shielded FFC
Battery drain	Won't last 48h	Proper sleep modes, kill peripheral power
Load cell drift	Readings vary with temp	Temperature compensation, periodic tare
Audio distortion	Clipping at high volume	Limit gain to 12dB, soft clipping
Optical miscounts	Ambient light false triggers	Shield sensor gap, check LED current

APPENDIX D

Debug Serial Commands

```
> status            System overview (WiFi, battery, temp, API)
> motor test 1      Test motor 1 rotation
> motor test all    Test all motors
> sensor test       Read all sensor values
> dispense 1 2      Dispense slot 1, 2 pills
> calibrate loadcell Load cell calibration
> i2c scan 0        Scan I2C bus 0 (touch)
```

```
> i2c scan 1        Scan I2C bus 1 (sensors)
> wifi status        WiFi connection details
> version            Firmware version & build date
> reboot             Restart device
```

APPENDIX E

Second-Source Strategy

COMPONENT	PRIMARY	SECOND SOURCE
MCU	ESP32-S3-WROOM-1-N16R8	ESP32-S3-WROOM-1-N8R8
Power Path	BQ24195	BQ25895
5V Buck	TPS62150	AP62150 (Diodes Inc)
3.3V LDO	AP2112K	XC6220B331MR
Motor Driver	ULN2003AN (TI)	ULN2003A (ST)
Temp Sensor	SHT40	SHT31
Optical	TCPT1300X01	GP1A57HRJ00F (Sharp)
Load Cell ADC	HX711	NAU7802